PLC STM32F407VET6 + W5500 : Modbus TCP (LAN) + Modbus RTU (RS485), nap code qua USB-C (CH340C)

Nguon 24VDC

File: power.kicad_sch

MCU STM32F407VET6

File: mcu.kicad_sch

USB nap code

File: usb.kicad_sch

Ethernet + RS485

File: comm.kicad_sch

16 ngo ra transistor

File: outputs.kicad_sch

10 ngo vao analog

File: analog.kicad_sch

Sheet: Root

PLC STM32F407 + W5500 Modbus TCP

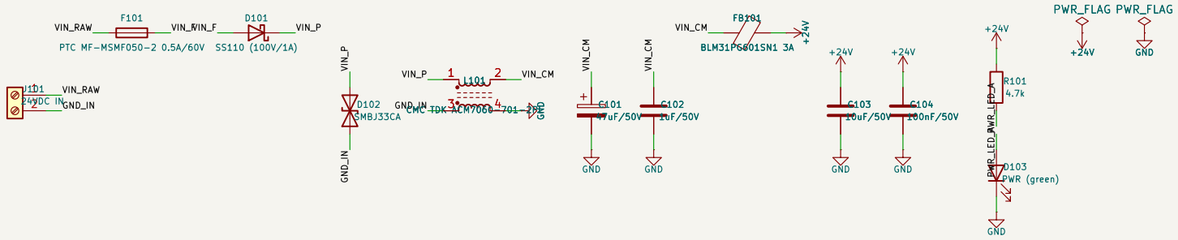
Sheet: /

File: PLC_STM32_Modbus.kicad_sch

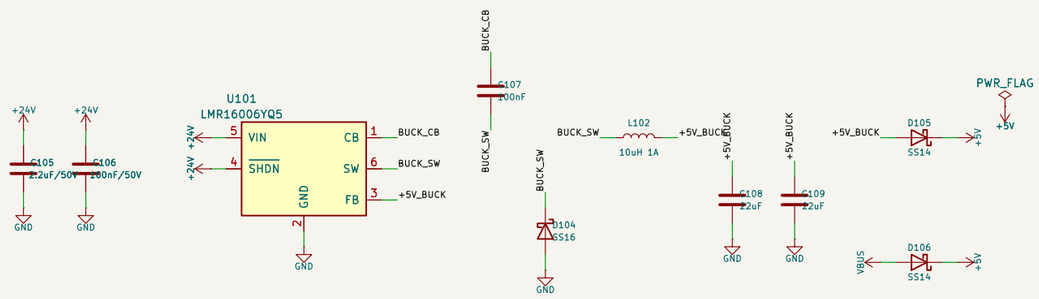
Title: PLC STM32F407 Modbus TCP – 16DO / 10AI

Size: A3	Date: 2026-10-14	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 1/7

NGUON VAO 24VDC: cau chi -> chong nguoc -> TVS -> loc common-mode -> loc LC -> +24V

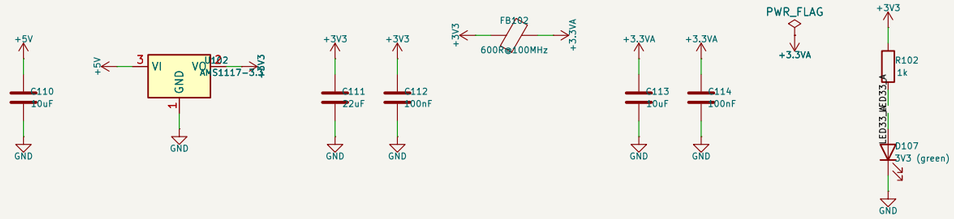


BUCK 24V -> 5V (LMR16006Y, 60V, 0.6A, 2.1MHz)



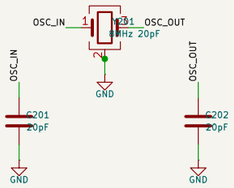
Cap nguon tu USB khi chua co 24V (de nap code)

LDO 5V -> 3.3V, loc rieng 3.3VA cho ADC

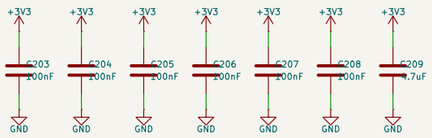


Sheet: Nguon 24VDC		
PLC STM32F407 + W5500 Modbus TCP		
Sheet: /Nguon 24VDC/		
File: power.kicad_sch		
Title: Nguon: loc 24VDC, buck 5V, LDO 3.3V, ref 3.0V		
Size: A3	Date: 2026-10-14	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 2/7

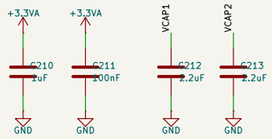
HSE 8MHz



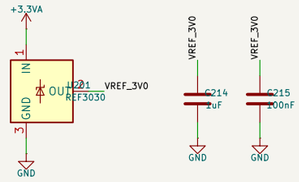
Tu loc nguon: 1 tu 100nF sat moi chan VDD



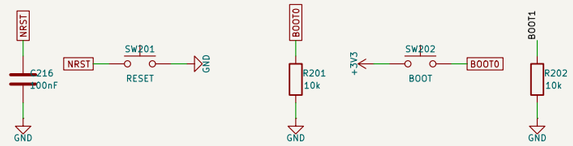
VDDA / VCAP

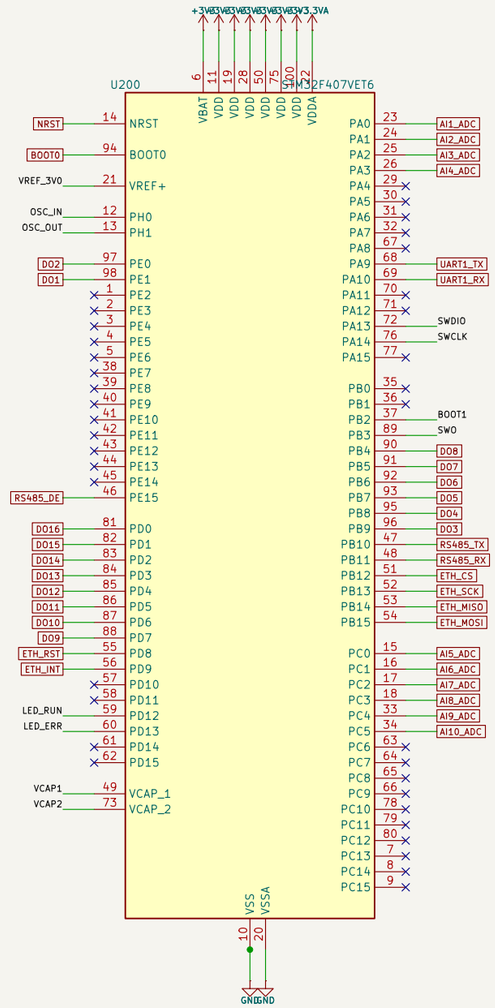


Dien ap chuan ADC 3.000V (VREF+)

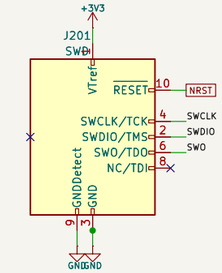


RESET / BOOT

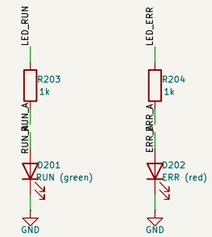




SWD debug (ST-Link)



LED trang thai



GAN CHAN:

DO1..DO16 -> PE1,PE0,PB9,PB8,PB7,PB6,PB5,PB4,PD7..PD0

PA0-PA3, PC0-PC5 -> AI1..AI10 (ADC1 12-bit)

SPI2 PB12(CS) PB13 PB14 PB15 -> W5500, PD8 RST, PD9 INT

USART1 PA9/PA10 -> CH340C (bootloader UART)

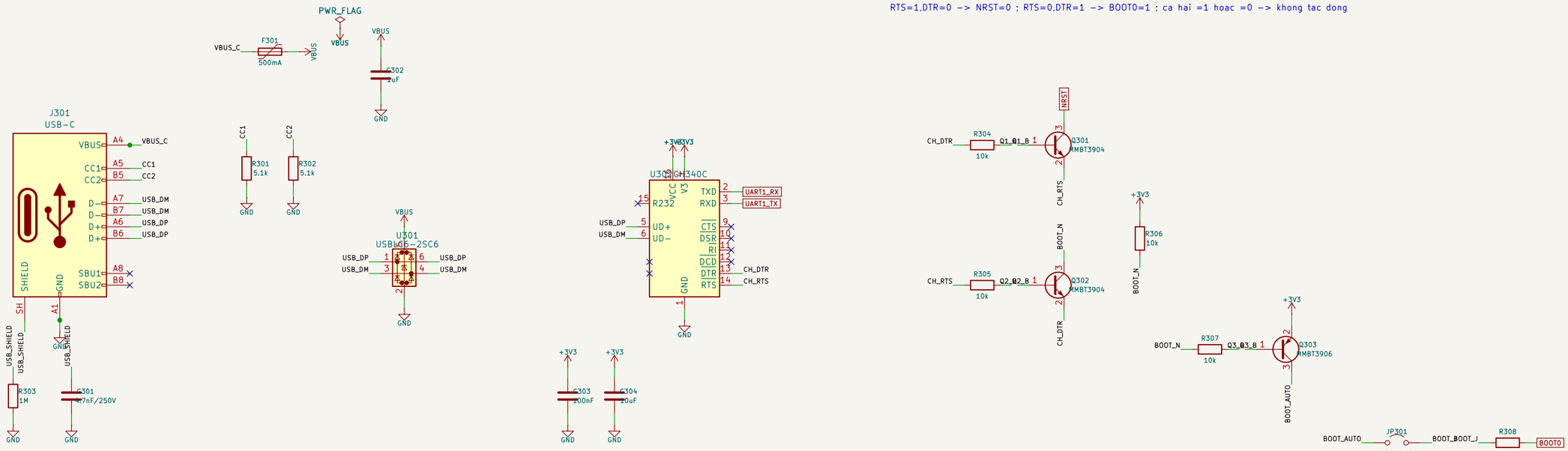
USART3 PB10/PB11 + PE15 DE -> RS485

PA13/PA14/PB3 -> SWD, PD12/PD13 -> LED

Sheet: MCU STM32F407VET6
PLC STM32F407 + W5500 Modbus TCP
Sheet: /MCU STM32F407VET6/
File: mcu.kicad_sch
Title: Vi dieu khien STM32F407VET6
Size: A3 | Date: 2026-10-14 | **Rev: 1.0**
KiCad E.D.A. 10.0.6 | Id: 3/7

MACH TU DONG NAP (DTR/RTS) – giong ESP32

RTS=1,DTR=0 -> NRST=0 ; RTS=0,DTR=1 -> BOOT0=1 ; ca hai =1 hoac =0 -> khong tac dong



Sheet: USB nap code

PLC STM32F407 + W5500 Modbus TCP

Sheet: /USB nap code/

File: usb.kicad_sch

Title: USB–C -> CH340C -> USART1 + mach tu dong vao bootloader

Size: A3	Date: 2026–10–14	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 4/7

Phoi hop tro khang PHY (theo WIZnet ref. schematic – kiem tra lai)

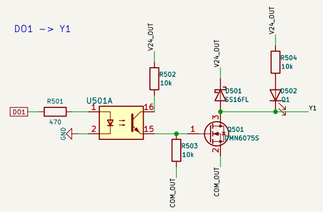
RS485 (Modbus RTU) – USART3

Tu loc W5500 (dat sat chan AVDD/VDD)

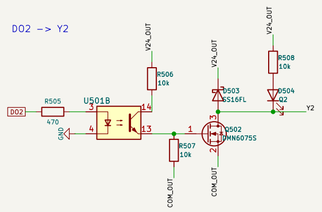
Sheet: Ethernet + RS485		
PLC STM32F407 + W5500 Modbus TCP		
Sheet: /Ethernet + RS485/		
File: comm.kicad_sch		
Title: W5500 Ethernet (Modbus TCP) + RS485 (Modbus RTU)		
Size: A3	Date: 2026-10-14	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 5/7

16 NGO RA TRANSISTOR SINK (NPN–type): MCU –> TLP291 –> MOSFET 60V/2.5A. Tai noi giua V+ (24V) va Yn. Cap nguon ngoai vao V+/COM (cach ly voi mach logic).

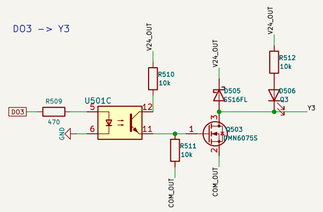
DO1 –> Y1



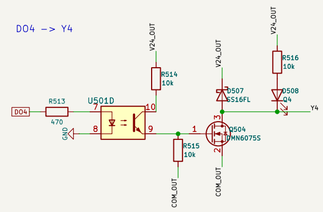
DO2 –> Y2



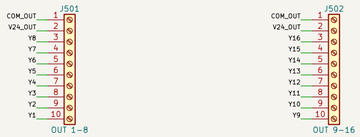
DO3 –> Y3



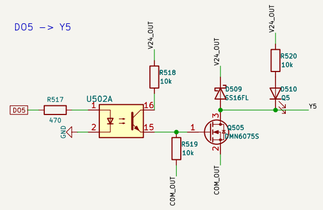
DO4 –> Y4



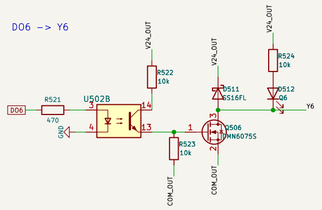
CAU DAU NGO RA



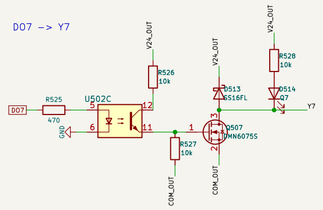
DO5 –> Y5



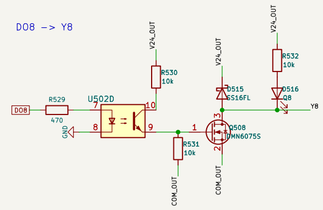
DO6 –> Y6



DO7 –> Y7



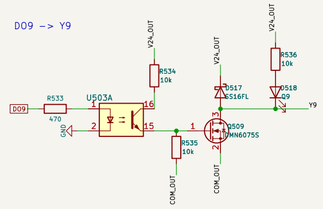
DO8 –> Y8



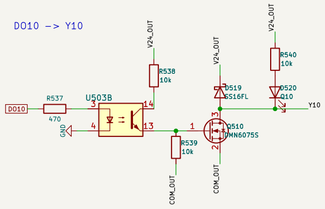
Loc / bao ve nguon ngo ra



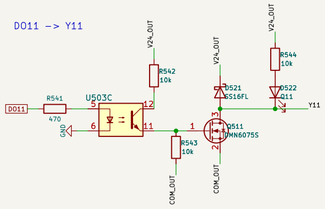
DO9 –> Y9



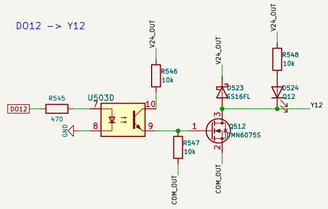
DO10 –> Y10



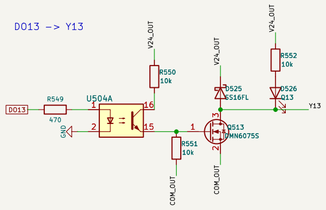
DO11 –> Y11



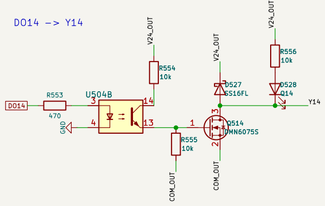
DO12 –> Y12



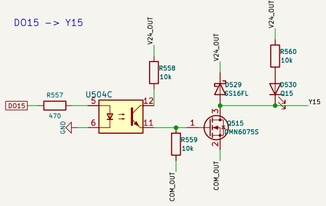
DO13 –> Y13



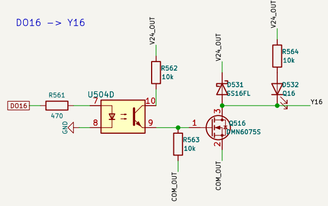
DO14 –> Y14



DO15 –> Y15



DO16 –> Y16



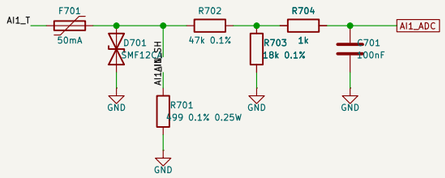
Sheet: 16 ngo ra transistor
PLC STM32F407 + W5500 Modbus TCP
Sheet: /16 ngo ra transistor/
File: outputs.kicad_sch

Title: 16 ngo ra transistor (sink), cach ly quang

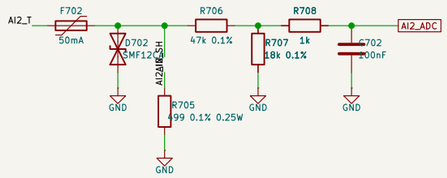
Size: A2	Date: 2026-10-14	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 6/7

10 NGO VAO ANALOG: 0–10V (ho JP) hoac 4–20mA (dong JP, shunt 499R). Vadc = Vin x 18/65 ; 10V –> 2.77V ; 20mA –> 2.74V (VREF = 3.000V)

AI1: 0–10V / 4–20mA (cam jumper 1 tren J703)



AI2: 0–10V / 4–20mA (cam jumper 2 tren J703)



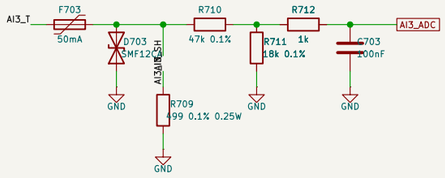
J701
AI 1–10

Nguon cap cho cam bien 2 day (24V loop) + chan chung GND

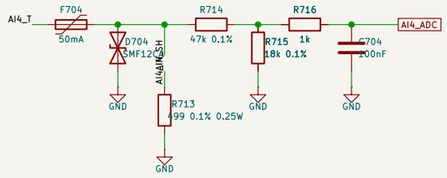


J703
AI PWR/COM

AI3: 0–10V / 4–20mA (cam jumper 3 tren J703)



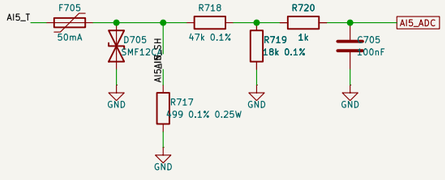
AI4: 0–10V / 4–20mA (cam jumper 4 tren J703)



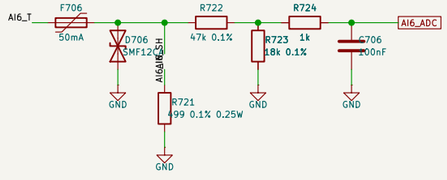
J702 mA SELECT

J703: cam jumper cap chan (2n–1,2n) –> kenh n do 4–20mA

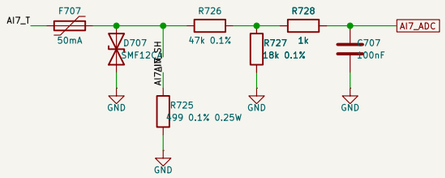
AI5: 0–10V / 4–20mA (cam jumper 5 tren J703)



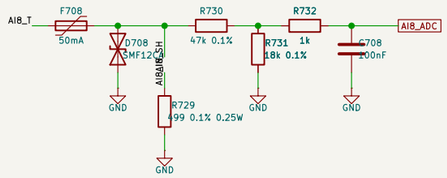
AI6: 0–10V / 4–20mA (cam jumper 6 tren J703)



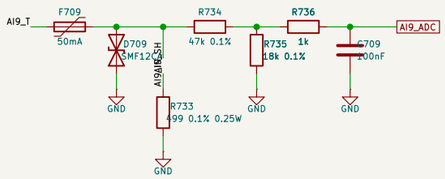
AI7: 0–10V / 4–20mA (cam jumper 7 tren J703)



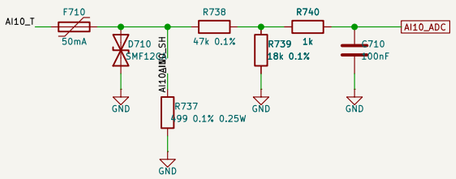
AI8: 0–10V / 4–20mA (cam jumper 8 tren J703)



AI9: 0–10V / 4–20mA (cam jumper 9 tren J703)



AI10: 0–10V / 4–20mA (cam jumper 10 tren J703)



Sheet: 10 ngo vao analog
PLC STM32F407 + W5500 Modbus TCP
Sheet: /10 ngo vao analog/
File: analog.kicad_sch
Title: 10 ngo vao analog 0–10V / 4–20mA
Size: A3 | Date: 2026–10–14 | **Rev: 1.0**
KiCad E.D.A. 10.0.6 | Id: 7/7